FLEXSIGHT

IoT Temperature Monitoring and Alert System

Holberton School — Final Portfolio Project

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Project Manager · Hardware & Sensor Integration

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SCM · Backend Developer

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QA · Frontend Developer · Documentation

Live project: https://flexsight.dev/links

GitHub: https://github.com/rabea14180-max/Portfolio-Project

THE PROBLEM

Unnoticed Temperature Changes Cause Damage

- Sensitive environments can be damaged by temperature increases that go unnoticed.
- Manual, periodic checks are slow and do not provide continuous visibility.
- FlexSight provides centralized hourly monitoring, stored readings, and alerts.

Target Environments

- **Warehouses**
- **Factories**
- **Server Rooms**
- **Data Centers**
- **Electrical Rooms**
- **Technical Rooms**
- **Offices**

MVP Concept

A focused web dashboard that turns raw sensor readings into continuous visibility and timely alerts — for the people responsible for protecting sensitive spaces.

PROCESS SUMMARY

Project Journey

1. Ideation & Planning
2. Charter & Scope
3. Technical Docs & System Design
4. MVP Development & Integration
5. Testing & Validation
6. Final Delivery & Presentation

Key Decisions

- Focus only on temperature monitoring
- Use a web dashboard
- Use PostgreSQL for structured data
- Use role-based permissions
- Keep the MVP focused and realistic

PROCESS SUMMARY

Development Across Three Sprints

Sprint 1

- Database structure
- Authentication
- Backend foundation
- Base React application

Sprint 2

- Dashboard pages
- REST APIs
- Devices, readings & alerts
- Initial sensor development

Sprint 3

- ESP32 integration
- Testing & bug fixing
- Deployment
- Final integration

16	3	5.3
Completed Tasks	Sprints	Avg. Velocity / Sprint

TECHNICAL DETAILS

Technologies and Technical Decisions

FRONTEND

React & Vite

Fast development with a component-based, maintainable UI

BACKEND

Python & Flask

Lightweight framework well suited to a focused REST API

DATABASE

PostgreSQL

Reliable, structured storage for devices, readings, alerts & users

HARDWARE

ESP32 & DHT11

Affordable, reliable temperature sensing with Wi-Fi connectivity

SOURCE CONTROL

Git & GitHub

Version control and collaboration across three team members

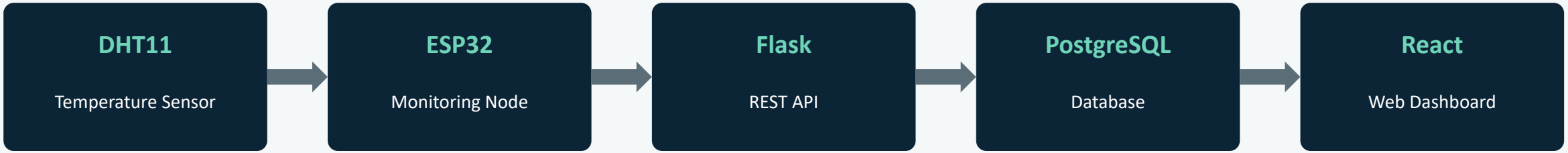
DEPLOYMENT

Cloud Deployment

Keeps the working MVP accessible for review and demo

TECHNICAL DETAILS

System Architecture



Captures live temperature readings

Sends readings to the Flask backend

Validates & processes readings

Stores devices, readings, alerts & users

Retrieves & displays data via REST APIs

How Alerts Work

The backend evaluates each incoming reading against a temperature threshold configured by an authorized user. When the configured threshold is exceeded, the system creates an alert inside the application, visible on the dashboard.

Core Features & Role-Based Access

Core Features

- Owner registration & secure login
- Dashboard with system statistics
- Device monitoring & status
- Stored, hourly temperature readings
- User-configurable temperature threshold
- Alert generation, severity & status
- Alert acknowledgement & resolution
- User management

Role-Based Access

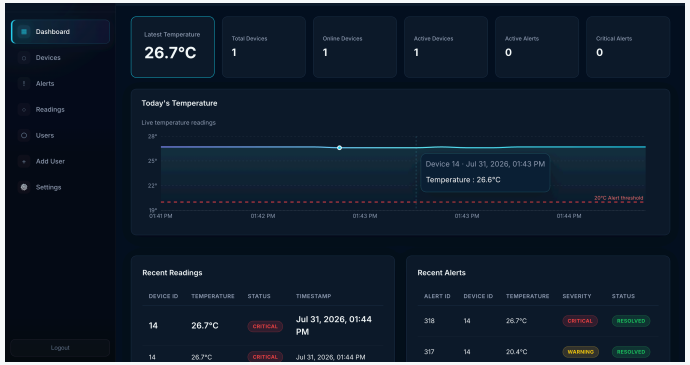
	Owner	Admin / Manager	Inspector
Dashboard	Full	Full	View only
Readings	Full	Full	View only
Users & Devices	Manage	Manage	—
Alerts & Threshold	Manage	Manage	—
Settings (password)	Full	Full	Change password

Inspector has view-only access to the Dashboard and Readings, and can use Settings only to change their password.

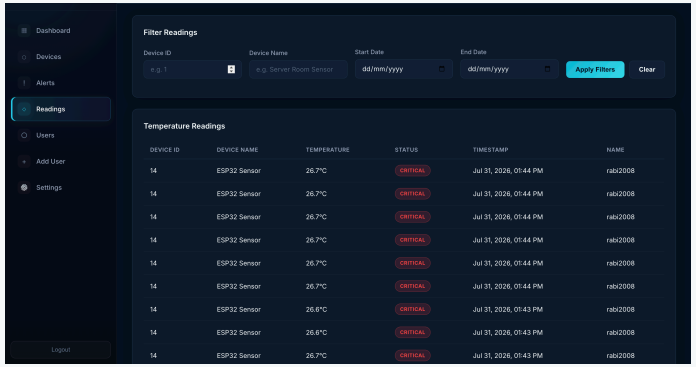
DEMO

Live MVP Demo

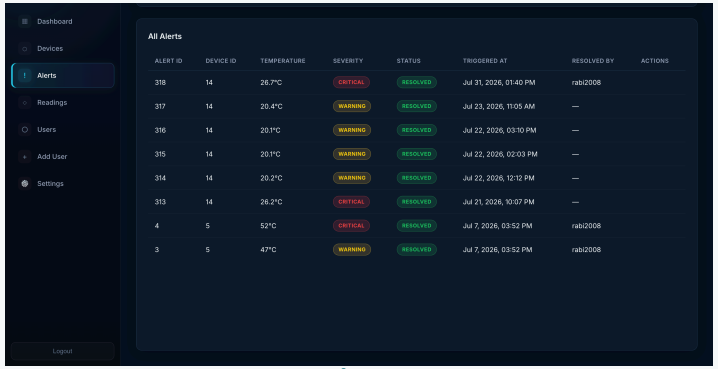
1. Open FlexSight and log in
2. Show the dashboard and system statistics
3. Show devices and device status
4. Show stored temperature readings
5. Change or explain the configurable threshold
6. Show generated alerts
7. Acknowledge or resolve an alert
8. Demonstrate role-based access



FlexSight Dashboard



Temperature Readings



Alerts

REFLECTION

Testing and Validation Results

38
Total Automated Checks Passed

26
Frontend Unit Tests

12
Backend & Integration Tests

Frontend Unit Test Breakdown

- **13** StatusBadge tests
- **4** ProtectedRoute tests
- **9** API Helper tests

Coverage Highlights

- Stable web MVP deployed
- Full sensor-to-dashboard workflow integrated
- API health, authentication & dashboard data validated
- Devices, readings, alerts & users covered
- Connectivity & production integration verified

REFLECTION

Challenges, Responses & Lessons Learned

Challenges

- Hardware and ESP32 integration
- Coordinating parallel frontend & backend work
- Testing the complete sensor-to-dashboard flow
- Keeping the MVP within the agreed scope

Responses

- Iterative sensor testing
- Clearer task ownership
- Incremental integration
- Early automated testing
- Focused scope decisions

Lessons Learned

- Begin hardware testing earlier
- Leave buffer time for integration
- Test small components before full integration
- Clear ownership improves team decisions

REFLECTION

Team Contributions & Retrospective

Hamsa Bnian Alammar

Project Manager

- Led planning & sprint coordination
- Worked on the ESP32 and sensor hardware
- Supported hardware integration

Rabea Thabit

SCM · Backend Developer

- Managed SCM and repository structure
- Built the Flask backend and REST APIs
- Implemented PostgreSQL integration

Munirah Alotaibi

QA · Frontend Developer

- Built frontend views and components
- Planned and executed QA testing
- Prepared and maintained project documentation

What Went Well

- All 16 tasks were completed
- Clear role distribution supported parallel progress
- Testing caught issues before final delivery

What Could Improve

- Earlier hardware testing
- More frequent cross-team integration checks

CONCLUSION

Conclusion & Next Steps

- FlexSight delivered a stable, deployed IoT temperature-monitoring MVP.
- The system connects a real sensor to a Flask backend, PostgreSQL database, and React dashboard.
- The implemented MVP meets its agreed scope.
- The system supports user-configured thresholds, alerts, and role-based access.

Possible Future Improvements

(not currently implemented)

- Mobile application
- Additional sensor types
- Advanced analytics and reporting
- Multi-branch support
- External system integrations

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Thank You